



Intelligent edge devices support complex protocols and packet encapsulations to manage diverse network topologies and deliver new services.

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Evolution on the Network Edge: Intelligent Devices

A network's edge is arguably one of its most important areas, because this is where it delivers services to network users—or, from the Internet service provider perspective, to subscribers. Devices at the network's edge classify, prioritize, and mark packets for the rest of the network to understand, and ultimately allow them into the network. Given modern networks' complex internetworking, it isn't surprising that developing and maintaining edge devices are tasks that head the agendas of most of today's network equipment manufacturers, carriers, service providers, and network administrators.

This article examines key elements of current network edge devices and the network topologies that work best with them. I also discuss Multiprotocol Label Switching (MPLS), a standardized solution that combines the performance and virtual circuit capabilities of data-link-layer switching with the proven scalability of network-layer routing used by today's edge devices. MPLS lets the edge device better prioritize Internet traffic and perform the core network routing more

efficiently, improve the quality of service (QoS), and provide traffic engineering functions that address the performance inconsistencies related to emerging Internet applications such as real-time voice and video streaming.

THE EDGE DEVICE

Cisco defines an edge device as

... a physical device that can pass packets between a legacy type of network such as an Ethernet network and an ATM [asynchronous transfer mode] network, using data link layer (OSI Layer 2) and network layer (OSI Layer 3) information. An edge device does not have responsibility for gathering network routing information, but simply uses the routing information it finds in the network layer using the route distribution protocol (*Internetworking Technologies Handbook*, third edition, Lowell Mauer and Greg Perry, Cisco Press, 2000).

For example, an *edge router* (sometimes called a boundary router) routes data between one or more local area networks (LANs) and an ATM backbone network—whether it is a campus network or wide-area network (WAN). In this case, the edge device has the difficult task of aggregating, forwarding, and routing traffic.

We can contrast the edge router with a *core router*, which forwards packets to computer hosts within a network—but not between networks. A core device or core router is part of the backbone, and it serves as the single pipe through which all traffic from peripheral networks must pass on its way to other peripheral networks. Unlike core routers, which primarily switch packets, edge devices not only handle basic IP (Internet Protocol)

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packet processing, but also consolidate multiple wire-speed traffic streams into flows with appropriate quality-of-service (QoS) parameters, perform traffic conditioning, and provide a multiplicity of services. Table 1 compares some of the characteristics and functions of core and edge devices.

Another example of an edge device is an *edge switch* such as Quarry Technologies' iQ8000 and iQ4000 Service Edge Switches. These operate in various edge networks that interconnect through a high-speed, high-capacity core network, which, in turn, connects to the outside world, as Figure 1 shows.

In addition to edge routers and switches, other types of edge devices include aggregation routers, and service delivery, next-generation voice, and multiservice provisioning platforms. Many edge devices, designed to bridge the gap between consumer demand and core capacity, play more than one of these roles. A single device might incorporate the functions of an aggregation router, IP service switch, and carrier-class router; serving all these functions would require high degrees of reliability, flexibility, scalability, and performance.

For example, to terminate and aggregate diverse traffic from the network, edge devices must support various interfaces and speeds, and they must also manage hundreds of thousands of individual concurrent sessions, each with a unique profile—T1/E1 channel groups, frame relay sessions, ATM permanent virtual circuits (PVCs), Ethernet virtual local area networks (VLANs), and so on.

NETWORK TOPOLOGIES

In considering how best to deploy edge devices, you should examine two network architecture types—centralized and distributed. From these two architecture types, network planners derive three major topological approaches: centralized, a mixture of centralized and distributed, and distributed.

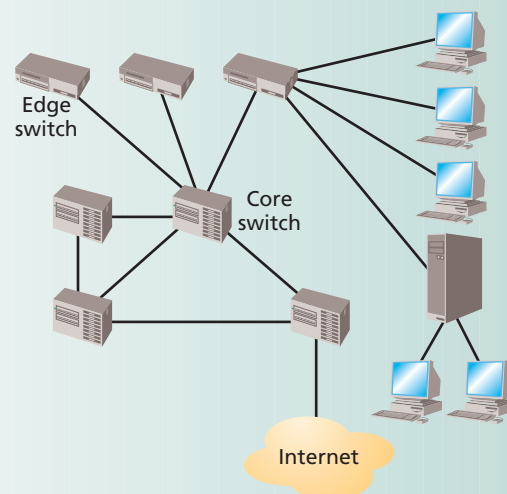
Centralized network

The centralized network topology assumes a *collapsed backbone*—a backbone network in which an intelligent edge device forwards frames while the central device analyzes each frame and makes decisions. Figure 2 shows an example of this topology. From the network management perspective, this approach is relatively easy to implement. However, it means dealing with frames at a very high line rate—10 to 100 times faster than with a distributed approach. The quantity of policies and flows that the backbone network must support is dramatically higher, mak-

Table 1. Core and edge device characteristics.

Device type	Throughput (Gbps)	Functions
Core	80 to 320	Packet forwarding Route processing
Edge	6 to 24	Packet forwarding Route processing Traffic shaping QoS scheduling Service features

Figure 1. Edge switch configuration device characteristics.



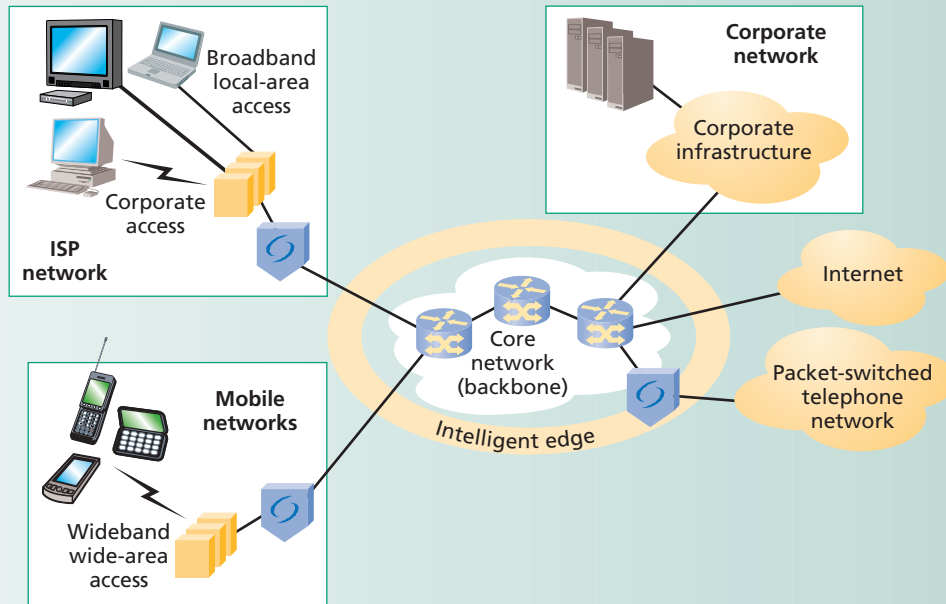
ing performance unpredictable. Lack of redundancy is another shortcoming of this approach; the central device becomes a single point of failure.

Combined centralized and distributed network

The combined centralized and distributed topology assumes that the edge device is not simple and can do some classification and policy treatment. Between the backbone node and the edge device, this topology uses a *middle device* that supports higher queue management and flow treatment levels. However, the combined approach suffers from multiple areas of weakness:

- It adds significant complexity and cost to the network without any appreciable gains. There are more nodes to install, manage, and support.

Figure 2. Centralized network example.



- The topology requires a new type of centralized device in the network architecture. This device has the same limitations as the central device in the centralized-network approach with no increase in benefits.
- Sharing QoS responsibilities between edge and middle devices that don't conform to the well-defined MPLS approach (Multiprotocol Label Switching, which this article describes in the next section) leads to configuration problems and inefficiency. Network effectiveness deteriorates because some fields of the packets must be inspected twice, and the network must implement the QoS and traffic standards twice.
- The middle device handles intensive queue management and flow-related activities. As a result, it adds latency and, sometimes, unpredictable jitter that adversely affects multimedia applications. Although we usually think of the Internet as introducing jitter and loss, even smaller, higher-bandwidth corporate networks can produce these effects. Much of the unpredictable packet loss and jitter in today's corporate networks trace back to how the best-effort-based internal-network routers deal with transient congestion. So the network's overall performance relates directly to the configurations of the internal core and edge devices and how the network incorporates routing policies and traffic shaping. Poor network performance can outweigh QoS facilities.

Distributed network

A distributed network that uses intelligent devices at the network edge is the preferable approach. These devices

perform frame analysis, classification, remarking, tagging, traffic shaping, and routing. They also perform analysis and decision making, though at relatively slower speeds.

A centralized network management device distributes the set of rules by which these devices operate. The central devices are *pipe only*, which means that they have high-performance switching and queuing capabilities but do limited analysis. For example, in an MPLS approach, edge devices do all the flow treatment and add or extract a *label*, a fixed-size field within the message header that corresponds to established paths through the network. The core devices simply perform label swapping and forward frames.

Although the centralized approach is easier to implement, the distributed network is more scalable, less sensitive to network failures, and has no single point of failure. It is more cost-effective and yields a more robust network with significantly better overall performance and efficiency.

MPLS ROUTING

The Internet Engineering Task Force (<http://www.ietf.org>) established the MPLS standards to achieve several capabilities, a few of which are

- fast forwarding speed;
- traffic engineering, including constraint-based routing, explicit routing, and the abilities to compute a path at the source, reserve network resources, and modify link attributes;
- support for voice and video on IP, including the ability to handle delay variation plus QoS constraints; and

Table 2. Edge device sampler: Vendors and products.

Vendor	Web address	Products	Web address
Cisco Systems	http://www.cisco.com	Cisco 10000 Series Routers	http://www.cisco.com/univercd/cc/td/doc/pcat/xtocid0
IP Infusion	http://www.ipinfusion.com	IP Routing/MPLS Software	http://www.ipinfusion.com/products/advanced/products_advanced.html
Juniper Networks	http://www.juniper.net	E-series for IP Services at the Edge	http://www.juniper.net/products/ip_infrastructure/e-series/erx_ip_services.html
Lucent Technologies	http://www.lucent.com	Spring Tide 7000 IP Service Switch Router and Wireless IP Service Switch Router	http://www.lucent.com/products/subcategory/0,,CTID+2017-STID+10024-LOCL+1,00.html
Marconi Communications	http://www.marconi.com	TNX-210 and TNX-1100	http://www.marconi.com/html/products/tnx210andtnx1100.htm
Network Appliance	http://www.netapp.com	NetCache C1200/C2100/C6100 Series	http://www.netapp.com/products/netcache/netcache_family.html
Nortel Networks	http://www.nortelnetworks.com	OPTera Metro 1000 series Ethernet Service Modules	http://www.nortelnetworks.com/products/01/optera/metro/1000
Quarry Technologies	http://www.quarrytech.com	IQ8000 and iQ4000 Service Edge Switches	http://www.quarrytech.com/products/iq_series_datasheet.shtml
Riverstone Networks	http://www.riverstonenet.com	Optical Metro Access Router RS 1000/3000	http://www.riverstonenet.com/products/router_rs1000.shtml

- support for virtual private networks, including the use of a controllable tunneling mechanism, equivalent to a frame relay or ATM virtual circuit (VC).

In an MPLS network, first an *edge-label-switching router* (edge LSR) assigns each incoming packet a label. Second, the router forwards the packet along a label-switched path, where each LSR makes forwarding decisions based solely on the label contents. Third, at each hop, the LSR strips the existing label and applies a new one that tells the next hop how to forward the packet. Finally, as the labeled packet leaves the network, another edge LSR removes the label.

For example, you can use MPLS to forward packets over the Internet backbone through the use of virtual private networks—through either border gateway protocol (BGP) or MPLS VPNs in either OSI layer 2 or layer 3. Used together, the MPLS signaling and VPN protocols establish a VPN tunnel, also known as label-switched paths, through which data travels. In addition, you can set up the MPLS label edge router to maintain multiple tunnels simultaneously.

CURRENT EDGE DEVICE USES

The evolving edge device is an extremely versatile and increasingly sophisticated creature. In its newest incarnations, this device supports several transport technologies and interface speeds to provide aggregation and connectivity among different network topologies and elements. Such devices also support an increasing number of complex protocols and packet encapsulations to manage diverse network topologies and deliver new services and applications. With the growing popularity of broadband services and third-generation, mobile wireless devices, tomorrow's edge devices will have to scale to manage even larger numbers of interconnected routers and available routes.

Economic and business forces, along with consumer demand for emerging Internet applications such as real-time voice and video streaming, have driven modern network architectures to a hierarchical structure with a fast, converged core and an intelligent edge. The functions of many different routers, switches, and aggregation multiplexers have coalesced in today's intelligent edge devices, which serve as Ethernet, frame relay, and ATM edge



More Reading on the Network Edge

- **“Deploying Service at the Network Edge,”** Karen Livoli, *EE Times*, 7 May 2002.
- ***Internetworking Technologies Handbook***, 3rd edition, Lowell Mauer and Greg Perry, Cisco Press, 2000.
- ***TCP/IP Protocol Suite***, Behrouz A. Forouzan, ed., McGraw-Hill, 2000.
- **“Powering Core Networks Through MPLS,”** Bilel Jamoussi and Tony Rybczynski, *Internet Telephony*, Nortel Networks, Jan. 2001.
- ***Implementing IP Services at the Network Edge***, David Ginsburg and Marie Hattar, Addison-Wesley, 2001.
- ***Quality of Service in IP Networks***, Grenville Armitage, Pearson Higher Education, 2000.

switches; IP service platforms; and carrier-class routers. Table 2 lists some commercially available edge devices.

Exchanging information between the network core and broadband access networks, these sophisticated edge devices simultaneously support widely diverse protocols, interfaces, and line rates. They handle heavy control- and data-plane loads. And they provide security, QoS, and many other IP services. Ideally, they can perform these

tasks without buffering IP packets for route lookup, and so operate at the speed of the underlying telecommunications technology—that is, at *wire speed*.

The network edge is no longer a simple demarcation point within a particular network; rather, it must contain a great deal of intelligence to meet the requirements of new technologies and of the business and regulatory environments. If customers are to receive reliable, affordable service—and if providers are to do business in a way that generates profit—network administrators need new tools to efficiently manage devices at the network edge.

Today, a new kind of IP network is evolving to meet the converging needs and technologies of several market segments—customers, ISPs, corporations, traditional telephony carriers, and other new types of customer-facing service providers. And the place where these market segments come together is the network’s edge. Understanding the important aspects of network topologies, edge devices, and MPLS routing can help you guide your own network’s evolution to participate effectively and profitably in this convergence. ■

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